

Python and High level
programming languages

High level programming languages

Scratch is an example of "high level programming languages"

so is python

high level just means that it's "removed" or "abstracted" from what the computer wants to do

remember that a computer can only do a very specific set of simple actions

this means that they're built with extra stuff to make it "easier"

this means that they don't work the way a computer actually works, but it's easier to read, has more convenience features, and has specific tools available that aren't how computers work but models how humans think well enough that it doesn't matter

this does mean it's slower, doesn't matter though

Why learn high level first

There are common worries for learning python first instead of something like c

there are a lot of arguments

1. not learning the fundamentals as well
2. slower language
3. builds habits specific to the language

doesn't matter

as long as you program a lot, you'll learn it anyway, i do recommend learning those at some point, i learned a bit of c, not good at it, but i understand it well enough

C

iconic, moisturized, the best in class, will never die

a bit hard to use, teaches really good fundamentals because basically every language is built on c

is the father of an entire style called c style languages

not actually the oldest or the fastest

Python

vaguely new, significantly easier to use, extremely popular

good for learning since they have lots of resources and it maps easier to english language

the slowness can suck for doing larger problems but scientific work does not use python anyway

it uses c under the hood of python so you can just sidestep that whole issue

Scratch

king of block games (other than unreal) disconnected completely from how computers work and prioritize human understanding over ease of computing

that's why its slow as hell

but it's so much easier

How programming languages work

what does the .txt mean

how do programming languages get installed

why do we need to install them

what happens when we install them

do we need to use the correct format?

shell

Thonny installation

why we're using thonny

portable so no installs

covers all our needs

allows for relatively easy plugin installatin

some nice learning features